

Why Does Capital Flow from Equal to Unequal Countries?

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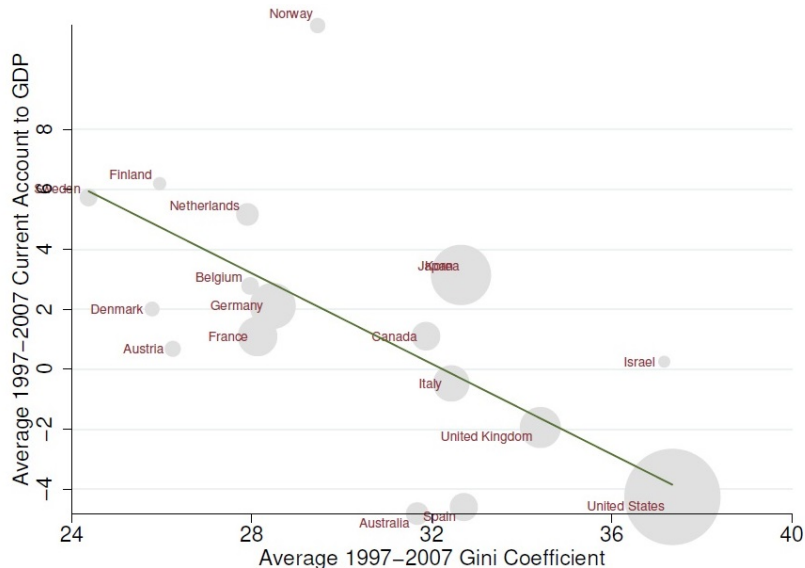
- What drives the current account imbalances?
- Show empirically that the current account balance is negatively correlated with the degree of income inequality
- Develop a theoretical model rationalizing this finding
 - Two countries
 - Heterogeneous agents
 - Endogenously incomplete financial markets

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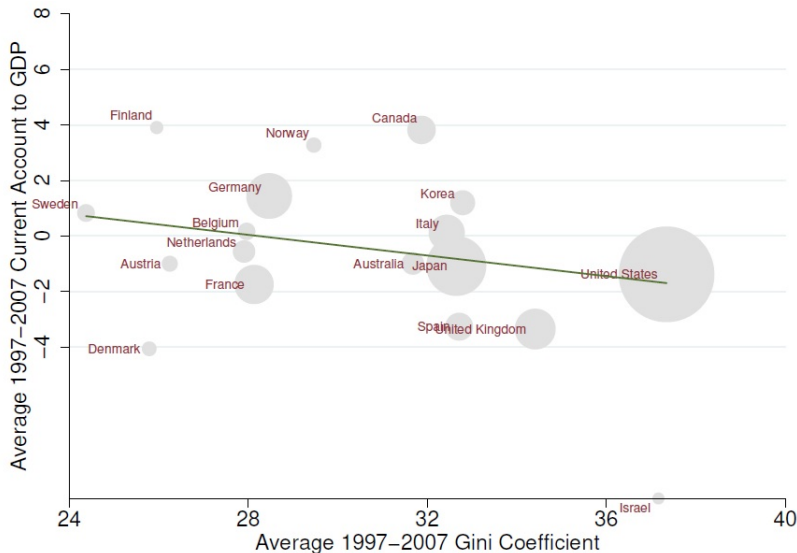
1 Empirical Findings

2 The Model

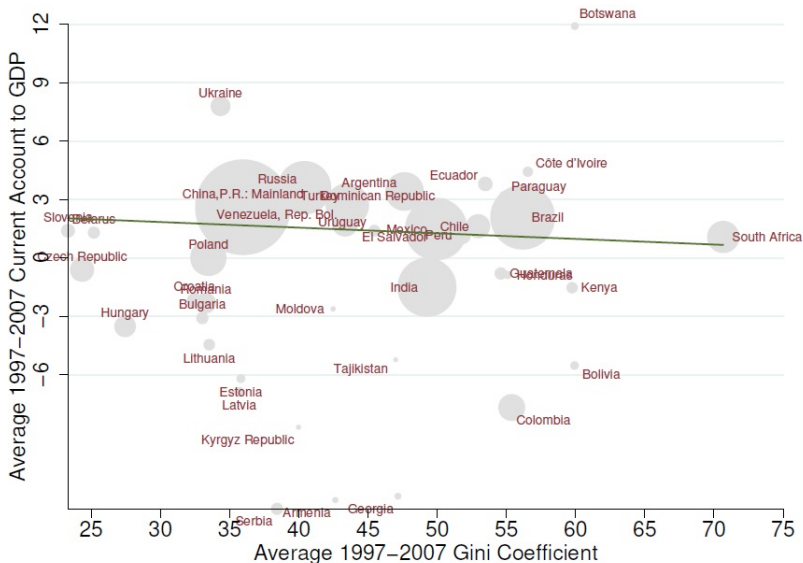
Income Inequality and the Current Account Balance in Advanced Economies



Income Inequality and the Current Account Balance in Advanced Economies with Controls



Income Inequality and the Current Account Balance in Emerging Economies with Controls



The Negative Correlation Is Driven by Private Savings

Table I: Relationship between Income Inequality and External Balance, Savings and Investment.^a

	Emerging	Advanced	Pooled
Current Account to GDP	-0.175 (0.132)	-0.803*** (0.142)	-0.310** (0.097)
Trade Balance to GDP	-0.170 (0.175)	-1.026*** (0.182)	-0.455*** (0.117)
Aggregate Savings to GDP	-0.833*** (0.267)	-1.296*** (0.235)	-0.985*** (0.164)
Investment to GDP	-0.649*** (0.109)	-0.270 (0.234)	-0.525** (0.142)
Private Savings to GDP	-0.774*** (0.133)	-1.145*** (0.162)	-0.854*** (0.169)
Public Savings to GDP	-0.044 (0.149)	-0.191 (0.131)	-0.140 (0.095)

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1 Empirical Findings

2 The Model

Environment

- Two countries, $i = 1, 2$, discrete time
- Two households of equal mass in both countries, $j \in \{H, L\}$
- Two realizations of endowment, $y_{j,i,t} \in \{y_{H,i}, y_{L,i}\}$
 - If one type gets y_H , the other must get y_L
 - $y_{H,i} = y(1 + \epsilon_i)$ and $y_{L,i} = y(1 - \epsilon_i)$
 - Total endowment in each country is y
 - π is the probability that the endowment next period will stay the same
- Let country 1 have low income inequality, $\epsilon_1 < \epsilon_2$
- Household preferences are

$$\sum_{t=0}^{\infty} \beta^t u(c_{j,i,t}),$$

where $u(\cdot)$ is CRRA

Closed-Economy Planner Problem

- In a symmetric allocation,

$$c_j = y(1 + \tilde{\epsilon}_j),$$

where $\tilde{\epsilon}_j = \tilde{\epsilon}_H = -\tilde{\epsilon}_L$ is the degree of risk-sharing

- The planner maximizes the sum of two households' welfare

$$\max_{\tilde{\epsilon}, \tilde{\epsilon}_H, \tilde{\epsilon}_L} \sum_{j \in \{H, L\}} V(\tilde{\epsilon}_j)$$

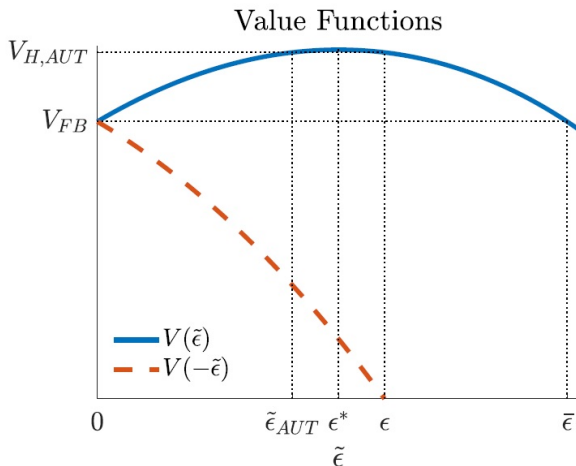
where $V(\tilde{\epsilon}_j)$ denotes the level of welfare attained by the household j

$$V(\tilde{\epsilon}_j) = u(c_j) + \beta [\pi V(\tilde{\epsilon}_j) + (1 - \pi) V(\tilde{\epsilon}_{-j})]$$

- Incentive compatibility constraint: a household can default and enter autarky

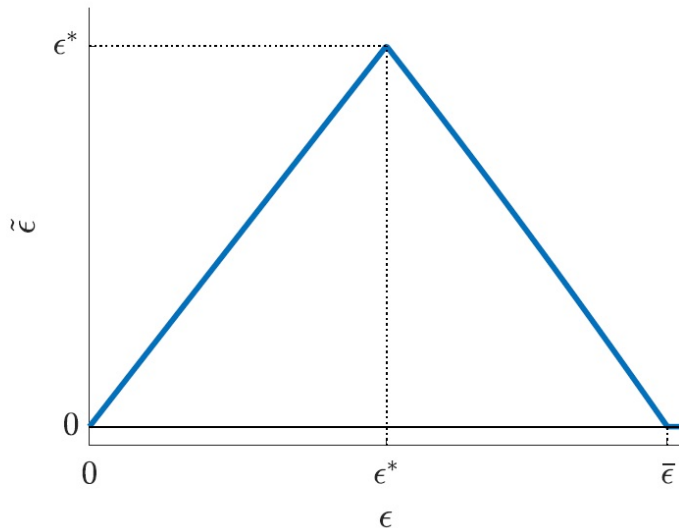
$$V(\tilde{\epsilon}_j) \geq V_{j,AUT} \quad \forall j \in \{H, L\}$$

Closed-Economy Planner Value Functions



- ϵ^* is the global maximum
- $\bar{\epsilon} > 0$ is the value equivalent full risk-sharing, $V(\bar{\epsilon}) = V(0)$
- $\tilde{\epsilon}_{H,AUT} < \epsilon^*$ is the lowest degree of risk-sharing equating the household value to the one in the autarky, $V(\tilde{\epsilon}_{H,AUT}) = V_{H,AUT}$

Closed-Economy Planner Solution



Closed-Economy Decentralized Allocation

- The equilibrium is identical to the planner allocation
- Households can issue one-period securities, and the budget constraint of a household j becomes

$$c_j = y_j - b_j + q_{j,H}b'_{j,H} + q_{j,L}b'_{j,L}$$

- Borrowing constraints are

$$b'_{j,H} \leq \bar{b}_H, b'_{j,L} \leq \bar{b}_L$$

- From market clearing

$$b'_{H,H} = -b'_{L,L}, b'_{H,L} = -b'_{L,H}$$

- In a stationary and symmetric allocation, $b'_{H,H} = b'_{L,H} = b_H$

Closed-Economy Decentralized Allocation: Solution

- Guess that the borrowing constraint never binds for high-endowment household
- Imposing the consumption from planner's allocation,

$$q_{H,H} = q_{L,L} = \beta\pi \text{ and } q_{H,L} = q_{L,H} = \beta(1 - \pi) \frac{u'(y(1 - \tilde{\epsilon}))}{u'(y(1 + \tilde{\epsilon}))}$$

- If full risk-sharing is not featured, $\tilde{\epsilon} > 0$, the constraint for the low type binds
- Can get solution as a function of exogenous and endogenous inequality

$$b_H = -b_L = -\bar{b}_L = \frac{y(\epsilon - \tilde{\epsilon})}{1 - q_{H,H} + q_{H,L}} = \frac{y(\epsilon - \tilde{\epsilon})}{1 - \beta \left[\pi - (1 - \pi) \frac{u'(y(1 - \tilde{\epsilon}))}{u'(y(1 + \tilde{\epsilon}))} \right]}$$

Open-Economy Planner Problem

- The planner maximizes the sum of the four households in the two countries

$$\max_{\tilde{\epsilon}, \tilde{\epsilon}_H, \tilde{\epsilon}_L, \tau, \tau_1, \tau_2} \sum_{i=1}^2 \sum_{j \in \{H, L\}} V(\tilde{\epsilon}_j, \tau_i)$$

- The planner chooses the transfers $\tau_1 = -\tau_2 = \tau$ across the two countries and sets the degree of consumption risk $\tilde{\epsilon}$
- Can interpret transfer as interest payments between the two countries on previously incurred debt
- Incentive compatibility constraint: any household can default and enter autarky

$$V(\tilde{\epsilon}_j, \tau_i) \geq V_{j,AUT,i} \quad \forall j \in \{H, L\}$$

- In a symmetric allocation,

$$c_{j,i} = y(1 + \tilde{\epsilon}_j)(1 + \tau_i)$$

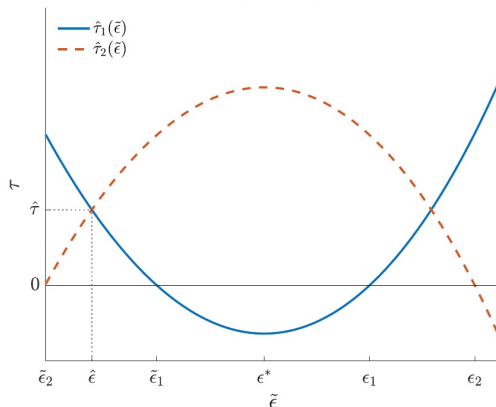
$$\tilde{\epsilon}_H = -\tilde{\epsilon}_L = \tilde{\epsilon}$$

Open Economy Planner Solution [1]

- If $\epsilon_2 > \epsilon_1 > \bar{\epsilon}$, the constraints don't bind, and the planner reaches the unconstrained optimum $\tilde{\epsilon} = \tau = 0$
- If both constraints bind, the solution is $(\hat{\epsilon}, \hat{\tau})$ s.t.

$$V(\hat{\epsilon}, \hat{\tau}) = V_{H,AUT,1} \quad \text{and} \quad V(\hat{\epsilon}, -\hat{\tau}) = V_{H,AUT,2}$$

Incentive Compatibility Constraints



Open Economy Planner Solution [2]

- If the closed-economy constraint is only binding in country 1, then (ϵ_1^*, τ_1^*) is a solution of a simplified problem, where IC for country 2 is omitted and in country 1 it holds with equality
 - If $\hat{\tau} > \tau_1^* > 0$, only IC in country 1 binds in the full problem, $\tau = \tau_1^*$. The consumption risk must be high
 - If $\tau_1^* > \hat{\tau} > 0$ and $\epsilon_1^* > \hat{\epsilon}$, IC in country 2 is violated and the planner sets $\tau = \hat{\tau}$
 - If both τ_1^* and $\hat{\tau}$ are negative, no constraint binds, the planner sets $\tau = 0$
- The solution of the full problem is

$$\tau = \max \{ \min \{ \tau_1^*, \hat{\tau} \}, 0 \}$$

Open-Economy Decentralized Allocation

- Market clearing becomes

$$\sum_{i=1,2} (b'_{H,H,i} + b'_{L,L,i}) = 0 \text{ and } \sum_{i=1,2} (b'_{H,L,i} + b'_{L,H,i}) = 0$$

- Net external debt positions are given by

$$b_{H,1} + b_{L,1} = -2y \frac{\tau}{1 - (q_{H,H} + q_{H,L})} = -b_{H,2} - b_{L,2}$$

- The asset positions are

$$b_{H,1} = \frac{y}{1 - q_{H,H} + q_{H,L}} \left[\epsilon - \tilde{\epsilon} - \tau \left(1 + \tilde{\epsilon} + 2 \frac{q_{H,L}}{1 - \tilde{q}} \right) \right]$$
$$b_{L,1} = \frac{y}{1 - q_{H,H} + q_{H,L}} \left[\tilde{\epsilon} - \epsilon - \tau \left(1 - \tilde{\epsilon} + 2 \frac{q_{H,L}}{1 - \tilde{q}} \right) \right]$$

Conclusions and Comments

- Document that for advanced economies, the current account to GDP ratio is negatively correlated with inequality
- Use a model with endogenous financial markets to rationalize this finding
- No quantitative model
- Why not applicable for emerging economies?
 - Individual fixed effects?
 - Persistence of shocks?